NCFE Level 2 Functional Skills Qualification in Mathematics (603/5060/X)

Paper number:	Practice P001270
Section B:	Calculator Test



Time allowed: 1 hour 30 minutes

Learner instructions

- Answer **all** questions.
- Read each question carefully.
- Write your answers in the spaces provided.
- Show your working, as marks may be awarded for working.
- State units in your answers, where appropriate.
- Check your work.
- Use $\pi = 3.14$

Learner information

- Section B contains **Activities 2, 3** and **4**.
- The maximum mark for this section is **45**.
- The marks available for **each** question are shown in brackets.

Resources

You will need a:

- pen, with black or blue ink
- pencil and eraser
- 30 cm ruler
- protractor
- calculator.

If extra pages are used, please make sure your name and centre name is on them and they are securely fastened to this booklet.

Please complete the details below clearly and in BLOCK CAPITALS.

Learner name _____

Centre name

Learner

All of the m

Practice Paper

OFFICIAL MFS MODEL SOLUTION · BENCHMARK EXEMPLARY PASS

Full Marks / Exemplary (100% Pass · Model Solutions)

· Assessment: Practice Paper 2 Section B

MASTER FUNCTIONAL SKILLS

ONLINE LEARNING PLATFORM & ACCREDITED COURSES

Pearson Edexcel & Open Awards Accredited Suite · Entry 3, Level 1 & Level 2

Pass Your Functional Skills Exams With Confidence

100% Pass Guarantee | Dyslexia-Friendly Learning | 1-to-1 Expert Tutor Support

Master Functional Skills provides an adaptive, dyslexia-friendly online learning platform tailored for adult learners, apprentices, and NHS healthcare staff. Everything you need to revise, practice, and pass is at your fingertips.

1. Smart Diagnostic Initial Assessment

Calculate your exact working level and pinpoint knowledge gaps instantly

- **Exact Grade Calculation:** Adaptive assessments calculate your exact level across all topics.
- **Personalised Study Pathway:** Skip what you know; focus 100% on the topics needed to pass.
- **Real-Time Exam Readiness:** Predictive tracker confirms when you are 100% ready to book.

2. Comprehensive Video Lessons & Worked Solutions

Bite-sized visual explainers covering every single Maths & English curriculum standard

- **Step-by-Step Video Lessons:** Master percentages, algebra, perimeter, ratio, and reverse fractions.
- **Multi-Ink Handwriting Models:** Visual 3-colour methods showing working, carries, and examiner tips.
- **Tutor Shortcuts & Golden Rules:** Proven non-calculator mental arithmetic hacks and BIDMAS tricks.

3. Full Mock Exams & Real-Time Performance Analytics

Practice in exam conditions and watch your scores and confidence climb

- **Authentic Exam Papers:** Timed Section A & Section B papers formatted exactly like Pearson Edexcel.
- **Instant Topic Breakdown:** View percentage scores across Number, Measure/Shape, and Data.
- **Historical Score Tracking:** Monitor past attempts to ensure continuous improvement before exam day.

Ready to Master Your Functional Skills & Pass First Time?

Join thousands of healthcare staff, apprentices, and adult learners across the UK.

Instant 24/7 access on desktop, tablet, and mobile with no software to install.

START LEARNING TODAY AT

MasterFunctionalSkills.co.uk

Examir

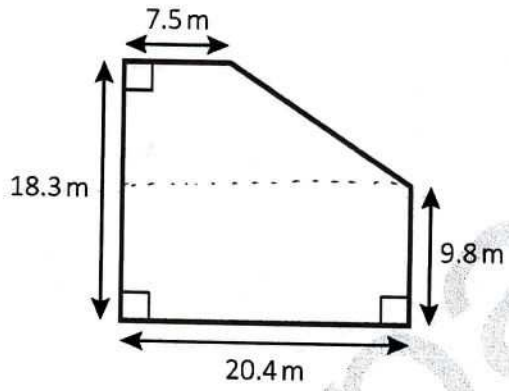
This page is intentionally left blank.

se only

2 (a) An allotment is a piece of land that people can rent to grow fruit, flowers and vegetables.

A residents' association has an allotment so that people can grow their own vegetables.

This is a plan of the allotment:



Not drawn accurately

What is the area of the allotment?

[3 marks]

20.4 × 9.8 = 199.92m².

½ × (20.4 + 7.5) × (18.3 − 9.8) = 118.575m²

199.92 + 118.575 =

Your answer: 318.495 **m²**

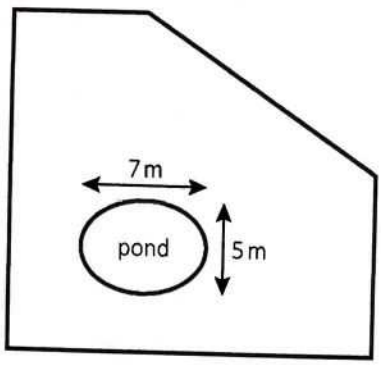
paper

Please turn over

nent.

The pond is 7 m long, 5 m wide and 80 cm deep.

The pond is to be filled in with soil.



Not drawn accurately

The formula to find the volume (in m^3) of the pond is: $\pi \times \frac{L}{2} \times \frac{W}{2} \times D$

Where:

L is the length (in m)
W is the width (in m)
D is the depth (in m)

Use $\pi = 3.14$

past paper

Soil can be bought in two sizes of bag.

Bag of soil	Volume of soil	Price per bag
A	0.9 m^3	£74
B	1.5 m^3	£110

The pond should be filled as cheaply as possible.

How much will it cost? Show your working.

[4 marks]

$3.14 \times \frac{7}{2} \times \frac{5}{2} \times 0.8 = 21.98\ m^3$

Bag A: $\frac{21.98}{0.9}$ ← round to 25 $\times$ £74 = [redacted] £1850

Bag B: $\frac{21.98}{1.5}$ ← round to 15 $\times$ £110 = [redacted] £1650

Use 15 bags of Bag B.

Your answer: £ 1650

Exam

ld take them 8 hours to fill in the pond

How long would it take 5 people working at the same rate?

[2 marks]

$3 \times 8 = 24$ hours total

$\frac{24}{5} = 4.8$ hours.

Your answer: 4.8 hours.

2 (d) Asif wants to grow potatoes.

The **GrowPotatoes** website gives these instructions for planting potatoes:

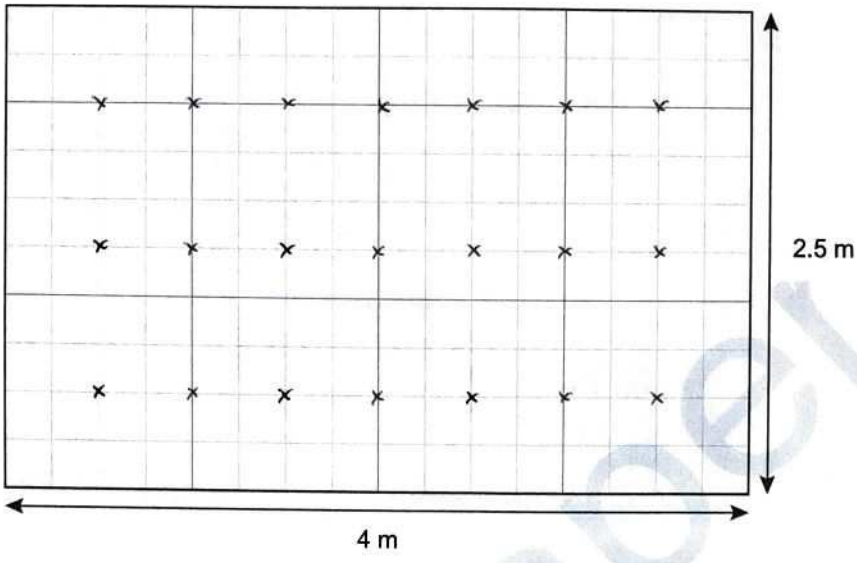
> Potatoes should be planted in rows.
>
> There must be 75 cm between the rows, and 50 cm between each plant in a row.
>
> There should be a gap of at least 50 cm between the plants and the edge of the plot.

The space for planting potatoes is a rectangle measuring 4 m by 2.5 m

Use the scale plan on the next page to show how the potato plants should be arranged to fit in the most potato plants.

Draw a cross (+) to mark the position for each plant.

[2 marks]



se only

past paper

Examir

…ut two types of potato, Cara and Des

	Yield (per potato plant)	
Trial number	**Cara**	**Desiree**
1	1.65 kg	1.25 kg
2	1.42 kg	1.42 kg
3	1.88 kg	2.10 kg
4	1.73 kg	1.76 kg
5	1.97 kg	

Yield is the weight of potatoes that can be produced from one potato plant.

Asif wants to plant the type of potato with the higher yield.
He will grow 20 potato plants.

Use the trial data to recommend which type of potato Asif should plant and estimate the expected total weight of potatoes, in pounds, that he will get.

Use the conversion: 1 kg = 2.2 pounds

[4 marks]

Mean:

Cara: $\frac{1.65+1.42+1.88+1.73+1.97}{5} = 1.73\text{kg}$

Desiree: $\frac{1.25+1.42+2.1+1.76}{4} = 1.6325\text{kg}.$

Median:

Cara: 1.73kg

Desiree: $\frac{1.42+1.76}{2} = 1.59\text{kg}.$

Use Cara potatoes

$20 \times 1.73 \times 2.2$

$= 76.12\text{lbs}$

Your answer:

Type of potato	Cara
	76.12 **pounds**

[Total marks: 15]

se only

3 (a) Pete is a catering student.
He has been given a project testing fruit juicers.

Pete is testing the SuperJuicer and the Blitz-It juicer.

Pete juices 100 g of strawberries in the Blitz-It juicer.
He repeats this 7 times.

Test number	Volume of juice
1	73 ml
2	66 ml
3	70 ml
4	65 ml
5	67 ml
6	71 ml
7	72 ml

Calculate the median volume of strawberry juice produced by the Blitz-It juicer.

[1 mark]

Your answer: 70 ml

Practice Paper

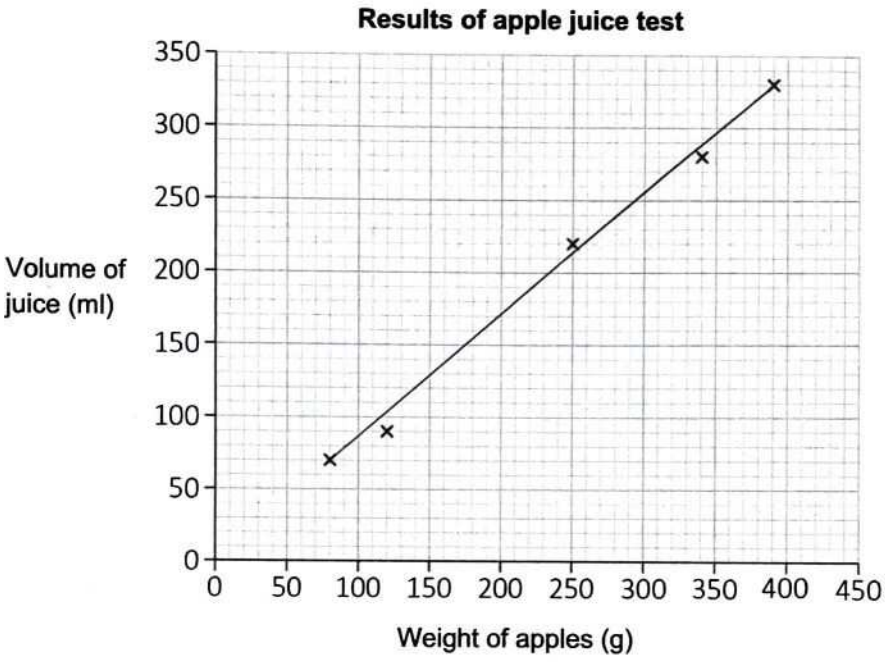
Please turn over

Exami

The manufacturer claims that 300 g of apples will produce $\frac{2}{5}$ of a pint of juice.

Pete tests this claim by juicing different weights of apples and measuring the amounts of juice they produce.

He uses his results to draw this scatter diagram:



' Show how you decide.

Use the conversion: 1 pint = 568 ml

[3 marks]

300g → 255 ml.

$\frac{255}{568} = 0.449$ pints.

$0.449 > \frac{2}{5}$.

Yes, the claim is fair, 300g produces at least $\frac{2}{5}$ of a pint.

Your answer: Yes.

past paper

Exam

s that the weight of juice produced from
t of the fruit.

If Pete wants to produce 150 g of juice, what weight of oranges should he use?

[2 marks]

$\frac{150g}{0.3} = 500g.$

Your answer: 500 g

3 (d) Pete asks a group of college students which machine's juice they prefer.

This table shows their responses:

Age group	SuperJuicer	Blitz-It juicer
Under 19 years old	16	4
19+ years old	17	13

A student from the group is chosen at random.

What is the probability that the student is under 19 and preferred juice from Blitz-It juicer?

[2 marks]

$16 + 4 + 17 + 13 = 50$

$4/50 = 0.08.$

Your answer: 0.08

g juice from pineapples and mango

He wants the volume of pineapple juice to mango juice to be as close as possible to a 3 : 2 ratio.

Typical volumes of juice produced from 1 fruit are:

Pineapple	0.2 litres
Mango	0.08 litres

Costs of fruit at the local supermarket are:

Type of fruit	Price for 1 fruit
Pineapple	£1.50
Mango	65p

What is the cost of the fruit that Pete needs to buy to make 8 litres of smoothie?

[5 marks]

3+2 = 5 parts. ⇒ $\frac{3}{5} \times 8L = 4.8L$ (pineapple) needed.

$\frac{2}{5} \times 8L = 3.2L$ (mango) needed.

$\frac{4.8}{0.2} = 24$ pineapples needed.

$\frac{3.2}{0.08} = 40$ mangoes needed.

$(24 \times £1.50) + (40 \times £0.65)$

$= £62$

Your answer: £ 62

use only

Past paper

Exan

...uit, he finds that pineapples have in
...n price to £1.75 each.

By what percentage has the price of pineapples increased?
Give your answer to 1 decimal place.

[2 marks]

$$\frac{£1.75 - £1.50}{£1.50} = 0.1\dot{6}$$

$$\Rightarrow 16.\dot{6}\ \%$$

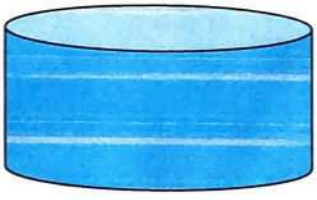
Your answer: 16.7 %

[Total marks: 15]

Past Paper

4 (a) Jan is an apprentice in a theatre.
She needs to plan the layout of the stage for the next play.

A prop is an object used on stage.
One of the props is a cylinder with radius 1.5 m



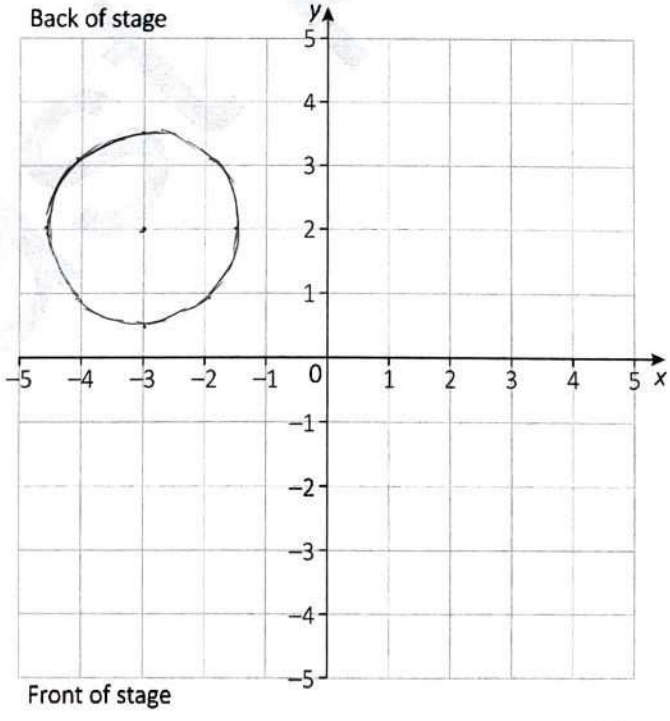
Jan uses a coordinate grid to plan where to put props on the stage.

Each small square on the grid represents a 1 m by 1 m square on the real stage.

She wants the centre of the cylinder to be at (–3, 2)

On the grid, draw a plan view of the cylinder in this position.

[2 marks]



Please turn over

d height 1.25 m

Calculate the volume of the cylinder in m^3
Give your answer to 1 decimal place.

Use $\pi = 3.14$

[2 marks]

$\pi \times 1.5^2 \times 1.25 = 8.83125\ m^3$

Your answer: 8.8 **m^3**

Past paper

tter.

The ball's radius is 60 cm

The formula to find the surface area (A) of a sphere is: $A = 4\pi r^2$

Jan buys the glitter in bottles.

She needs to use 300 g of glitter per square metre.

Each bottle holds 254 g of glitter.

Calculate the number of bottles of glitter needed to cover the sphere.
Show your working.

Use $\pi = 3.14$

[4 marks]

60 cm = 0.6m.

$A = 4 \times 3.14 \times 0.6^2 = 4.5216 m^2$.

$4.5216 \times 300g = 1356.48g$ glitter needed.

$\frac{1356.48}{254} = 5.34$ bottles → 6 bottles needed

Your answer: 6 **bottles**

ise only

Please turn over

Exami

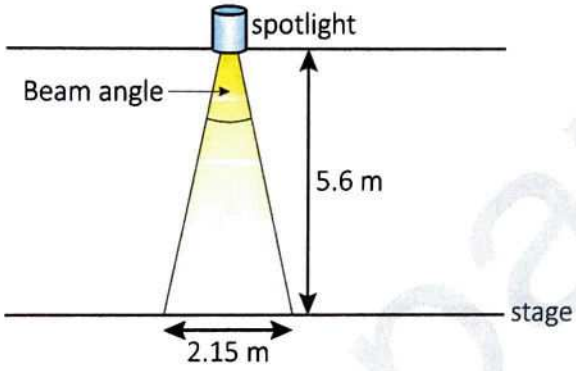
tlight that will give a circle of light on t
with a diameter of approximately 2.15 m

The formula for working out the diameter of the light on the stage is:

$$D = \frac{B \times H \times 18}{1000}$$

Where:

D is the diameter of the circle of light (in m)
B is the beam angle (in degrees)
H is the vertical distance from the spotlight to the stage (in m)



Past paper

spotlights:

Spotlight	Beam angle
P	19°
Q	26°

Which spotlight should be chosen to get the diameter closest to 2.15 m?
Explain how you decide.

[3 marks]

P: $\frac{19 \times 5.6 \times 18}{1000} = 1.9152\text{m}$.

Q: $\frac{26 \times 5.6 \times 18}{1000} = 2.6208\text{m}$.

P: $2.15 - 1.9152 = 0.2348\text{m}$ away

Q: $2.6208 - 2.15 = 0.4708\text{m}$ away.

Choose P.

Your answer: P.

Past paper

Exam

ains.

Jan has been asked to order the fabric for the new curtains.

The total width of fabric needs to be $3\frac{1}{5}$ times the width of the stage front.

Write $3\frac{1}{5}$ as a decimal number.

[1 mark]

Your answer: 3.2

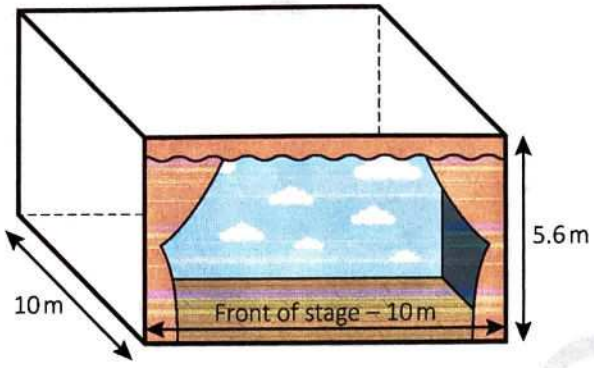
past paper

se only

This page is intentionally left blank.

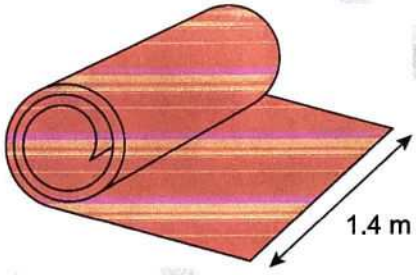
Please turn over for the next question.

past paper



The total width of fabric needs to be $3\frac{1}{5}$ the width of the stage front.

Curtain fabric is sold in widths of 1.4 m



Past Paper

together to make the curtains.

The length of the curtains needs to be 5.8 m to allow for hems.

What length of fabric does she need?

[3 marks]

$10 \times 3.2 = 32\text{m}$ (width needed)

$\frac{32\text{m}}{1.4\text{m}} = 22.86 \Rightarrow 23$ needed.

$23 \times 5.8\text{m} = 133.4\text{m}$

Your answer: 133.4 m

[Total marks: 15]

This is the end of the assessment.

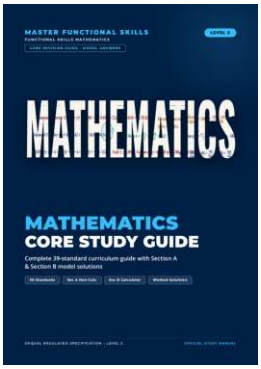
past paper

MASTER FUNCTIONAL SKILLS STORE

OFFICIAL REVISION GUIDES, WORKBOOKS & FLASHCARDS

Written by accredited specialist tutors to match the exact 2026 Pearson Edexcel standards

Everything You Need To Revise, Practice & Pass Your Maths Level 2 Exam



Core Curriculum Manual

Maths Level 2
Core Study Guide

- Full official curriculum coverage
- Section A Non-Calc & Sec B Calc
- Worked models & golden rules
- Dyslexia-friendly visual format

Spiral Bound Print & PDF

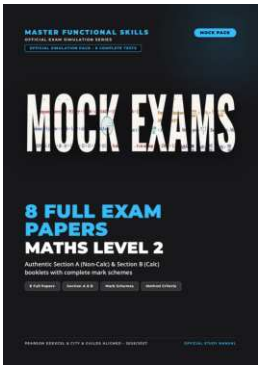


1,000+ Practice Drills

Maths Level 2
Companion Workbook

- 1,000+ exam-style questions
- Progressive difficulty drills
- Full worked step-by-step key
- Targeted fix for weak areas

Spiral Bound Print & PDF



Realistic Mock Papers

Maths Level 2
Official Mock Packs

- Full mock exam paper sets
- Authentic Pearson Edexcel style
- Detailed mark schemes included
- Grade boundaries & pass marks

Exam Booklet Pack & PDF



Pocket Flashcards A6

Maths Level 2
Pocket Revision Cards

- Compact A6 cards for quick study
- Essential formulas & conversions
- Dual-sided question & answer
- High-yield mental arithmetic rules

Boxed Card Deck & Print PDF

Order Official Study Guides & Revision Cards Online

Free Next-Day UK Tracked Shipping · Instant PDF Downloads Included

Trusted by NHS trusts, apprentices & adult learners nationwide.

VISIT THE ONLINE STORE

MasterFunctionalSkills.co.uk